

Spatial Audio Feedback for Discrete Rotation Technique in Virtual Reality

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Figure 1: Top-down view of the two audio source anchoring designs for discrete rotation in VR. The curved arrow shows a snap turn. In the *Fixed* condition (left), the beacon remains stationary in world space – after each snap turn, its perceived direction shifts relative to the listener, providing an immediate allocentric heading cue. In the *Moving* condition (right), the beacon repositions to follow the participant after each snap turn, eliminating any external spatial reference. Each anchoring condition is further combined with two pitch strategies – *Constant* and *Increasing* – where increasing pitch encodes rotational magnitude as a temporal frequency contour. Our study reveals that spatial stability combined with increasing pitch best supports spatial orientation under discrete rotation.

Abstract

Discrete rotation techniques such as snap turns are widely adopted in virtual reality (VR) to mitigate cybersickness by eliminating optical flow during sudden viewpoint changes. However, by collapsing rotational motion into instantaneous viewpoint updates, snap turns disrupt the perceptual signals that support spatial updating, fragmenting users' internal spatial representations. This paper investigates whether spatial audio feedback can support spatial



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orientation under discrete rotation. In a user study, we examine how two design components — audio source movement and pitch modulation — affect spatial orientation during snap turns. We varied each component across two levels (fixed vs. following beacon; constant vs. increasing pitch), yielding four conditions compared against a no-audio baseline. Results showed that fixed spatial audio with increasing pitch significantly improves spatial orientation accuracy, outperforming all other conditions including the no-audio baseline. These findings suggest that temporal auditory encoding can support spatial updating under discrete transformations, recommending that spatial audio feedback for discrete rotation should anchor the sound source to the environment and modulate pitch in proportion to rotation angle, preserving users' sense of directional continuity across discrete viewpoint changes.

CCS Concepts

• Human-centered computing → Virtual reality.

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1 Introduction

Although virtual reality (VR) has seen widespread adoption over the past decade, user comfort remains a major barrier to its further advancement because many users experience cybersickness, including nausea and disorientation. One widely deployed mitigation strategy is discrete rotation, e.g., snap turning, which reorients the user's viewpoint in fixed angular increments (typically 22.5°–45°) to reduce visually induced discomfort [Farmani and Teather 2018].

While snap turning can reduce cybersickness [Farmani and Teather 2020], it disrupts continuous visual motion cues that support spatial updating, making it harder for users to track their heading and position. Prior work on discrete and teleportation-based locomotion shows that interfaces lacking continuous self-motion or orientation cues can lead to larger orientation errors and slower task performance, highlighting an orientation maintenance cost of these techniques [Benda et al. 2023; Caputo et al. 2023; Farmani and Teather 2020].

Another potential approach to addressing this disorientation is spatial audio. Spatial audio provides directional sound cues that remain anchored to stable locations in the virtual environment; as the user's head orientation changes, these auditory cues update continuously, preserving information about heading and relative direction even when visual rotation occurs instantaneously. Unlike visual motion cues, auditory landmarks can persist across snap turns, suggesting a mechanism for supporting spatial updating without reintroducing visually induced cybersickness [Shayman et al. 2024b; Walker and Lindsay 2006].

Prior work has demonstrated that spatial auditory cues and sonified navigation instructions can improve wayfinding performance and spatial updating in virtual and real environments [Ahmetovic et al. 2023; Shayman et al. 2024b; Walker and Lindsay 2006]. However, existing work has not directly examined whether spatial audio

can mitigate the spatial disorientation introduced specifically by discrete rotation in VR. We address this gap by investigating spatial audio feedback for orientation maintenance during discrete rotation. In this work, we contribute to the growing body of research on multimodal feedback in VR by:

- **Identifying** that audio source anchoring and pitch modulation operate under distinct perceptual cues under discrete rotation: stable sound source gives users a fixed directional reference, and pitch contour encodes rotational magnitude.
- **Providing empirical evidence** that spatial stability is the necessary condition for auditory spatial updating: a world-fixed beacon supports orientation, whereas a beacon that follows the user does not, and may interfere with rather than support spatial updating.
- **Revealing** that pitch modulation serves as a temporal code for rotational magnitude, but at a measurable attentional cost — establishing an accuracy-comfort tradeoff with implications for designing auditory feedback in discrete locomotion interfaces.

2 Related Works

We review prior work on the following three topics: discrete rotation techniques and their effects on cybersickness, spatial orientation in virtual environments, and spatial audio as a feedback mechanism for supporting orientation during navigation.

2.1 Discrete Rotation Techniques in VR

VR users frequently experience cybersickness during continuous rotational movements due to the sensory conflict between visual motion and stationary vestibular states [Rebenitsch and Owen 2016]. Snap turning mitigates this conflict by replacing continuous rotation with instantaneous viewpoint updates, reducing SSQ scores by roughly 40% compared to continuous rotation [Farmani and Teather 2020], and continuous locomotion provoking significantly higher cybersickness than discrete methods [Hořejší et al. 2025]. Visual interventions such as dynamic FOV restrictors [Fernandes and Feiner 2016] and peripheral rest frames [Nie et al. 2025] have been explored to further reduce discomfort, but combining these with snap turning yields no additional benefit and can worsen experience [Kelly et al. 2022], suggesting visual-only mitigation has reached an efficacy ceiling.

Snap turning's comfort gain comes at a perceptual cost: by eliminating optic flow, it severs two signals the spatial updating system normally integrates, a continuous angular-magnitude signal and an environment-anchored reference frame [Buttussi and Chittaro 2023; Caputo et al. 2023]. This creates a comfort-orientation trade-off that visual approaches alone have not resolved, motivating cross-modal solutions. Whether any non-visual modality can restore these signals without reintroducing cybersickness is the open question we address.

2.2 Restoring Continuity in Discrete Locomotion

Researchers have attempted to restore spatial continuity through visual transition cues such as traceable teleportation [Jia et al. 2025] and optical flow during teleportation transitions [Bhandari et al.

2018], both of which improve spatial judgments but address translational rather than rotational discontinuities, leaving the angular-magnitude signal and allocentric reference frame disrupted by snap turns unrestored.

In redirected walking, rotational continuity is managed through gain controllers [Li et al. 2025; Mayor et al. 2022; Mizutani et al. 2023]. Junker et al. [Junker et al. 2021] found that spatialized audio does not affect detection thresholds for visual rotation gains, and Gerritse et al. [Gerritse et al. 2024] showed that spatial audio significantly increases curvature gain thresholds. These findings concern continuous, imperceptible rotation manipulation rather than the instantaneous viewpoint jumps of snap turning, where the user must reconstruct heading from a sudden scene change with no transitional information. Whether spatial audio can restore either severed signal across discrete rotation events remains unexplored.

2.3 Spatial Audio and Multimodal Feedback for Navigation

Spatial audio enhances immersion, presence, and orientation in virtual and real-world environments [Mavridou et al. 2025], with non-speech spatialized cues proving effective for guiding navigation and rotational alignment [Ahmetovic et al. 2023; Degara et al. 2014; Walker and Lindsay 2006]. However, users rely predominantly on visual cues for spatial updating and do not naturally shift to auditory information when visual input is disrupted [Shayman et al. 2024a], suggesting that spatial audio must be deliberately designed rather than passively available. Audiovisual landmarks improve spatial memory over unimodal cues [Seminati et al. 2022; Werkhoven et al. 2014], and design frameworks for auditory spatial representation [Guerreiro et al. 2023] and optimal audio placement [Cho et al. 2024] provide principled approaches for integrating audio into virtual environments.

Critically, prior work conflates two perceptual functions: providing an environment-anchored reference frame and directing the listener toward a navigation goal. Fixed beacons reduce angular pointing error in VR homing tasks [Shayman et al. 2024b] and outperform verbal instructions for reducing route deviations in real-world navigation [Gordon et al. 2023; Lystbæk et al. 2026], yet these studies do not isolate whether the benefit derives from spatial stability or directional guidance. For users with low vision, audio supports accessibility despite higher error rates than vision [Zhao et al. 2020]. Our study dissociates these functions by manipulating beacon anchoring and pitch contour orthogonally in a task where the beacon marks no navigation goal. No prior work has directly examined whether spatial audio can restore the allocentric reference frame or re-encode angular magnitude under discrete rotational updates — the gap this paper addresses.

3 Spatial Audio Feedback for Discrete Rotation in VR

We designed spatial audio feedback to support spatial orientation during discrete rotation by targeting two perceptual signals that snap turns disrupt. First, snap turns remove the environment-anchored reference frame that supports self-rotation registration — motivating **audio source anchoring** (Figure 1), which determines whether the beacon maintains a stable world-fixed position

or repositions to follow the user after each viewpoint update. Second, snap turns are instantaneous and provide no continuous signal of rotational magnitude — motivating **pitch modulation**, which examines whether a temporal pitch contour can serve as an auditory marker of the rotation event, consistent with established sonification principles for conveying scalar quantities through dynamic auditory contours [Dubus and Bresin 2013]. Together, these two components define a 2×2 design space of spatial audio feedback designs for discrete rotation.

3.1 Audio Source Anchoring

Fixed Beacon: The audio beacon was fixed in world space, emitting brief, spatially-anchored sound pulses that provided directional feedback relative to the participant’s current orientation. When participants performed snap turns, the perceived direction of the beacon shifted without the beacon physically moving, providing an immediate heading cue after each discrete viewpoint update [Shayman et al. 2024b; Walker and Lindsay 2006].

Moving Beacon: The audio beacon repositioned following each snap turn, orbiting the participant along a circular arc toward its new target position, maintaining a constant angular relationship to the listener. This design was motivated by prior work coupling feedback to locomotion events — using audio to signal the direction of a rotational viewpoint change [Lee et al. 2023] and haptic feedback to signal the magnitude of a locomotion event [Chu et al. 2021] — extended here to examine whether event-coupled audio supports spatial orientation in the absence of an allocentric reference. The arc movement duration was computed as:

$$\text{duration} = \frac{\text{arc length}}{\text{snapFollowSpeed}} = \frac{|\Delta\theta| \times r}{\text{snapFollowSpeed}} \quad (1)$$

where $\Delta\theta$ is the angular difference across the arc and r is the beacon orbital radius. Audio played continuously throughout the arc.

3.2 Pitch Modulation

Constant Pitch: The beacon emitted audio at a constant frequency for the duration of the rotation transition, providing stable directional information without temporal frequency change.

Increasing Pitch: The beacon emitted audio whose pitch dynamically ramped upward during the rotation transition, creating an upward frequency sweep synchronized with the snap turn. This temporal pitch progression was intended to encode rotational progression, providing additional information about angular displacement. The pitch ramp followed linear interpolation:

$$\text{pitch}(t) = \text{pitchStart} + t_{01} \times (\text{pitchEnd} - \text{pitchStart}) \quad (2)$$

where $t_{01} \in [0, 1]$ represents normalized progress through the transition.

3.3 Spatial Audio Feedback Designs

Combining the two types of **Audio Source Anchoring** (*Fixed* vs. *Moving*) and two types of **pitch modulation** (*Constant* vs. *Increasing*) yields four spatial audio feedback designs — *Fixed + Constant*, *Fixed + Increasing*, *Moving + Constant*, and *Moving + Increasing* — along with a *No-Audio* baseline, resulting in five designs total. Between rotations, the environment remained silent across all designs, ensuring that audio cues specifically supported spatial

updating during discrete rotation transitions rather than providing continuous background sound. We evaluated these five designs in a controlled user study to assess their effect on spatial orientation during discrete rotation.

4 User Study

We conducted a within-subjects study to evaluate the effect of spatial audio feedback on spatial orientation during discrete rotation, examining how two design components — **Audio Source Anchoring** and **Pitch Modulation** — independently and jointly affect orientation accuracy and perceived workload. Condition order was randomized independently for each participant to mitigate order effects.

4.1 Research Questions

- **RQ1:** Does **Audio Source Anchoring** affect spatial orientation accuracy during discrete rotation?
- **RQ2:** Does **Pitch Modulation** affect spatial orientation accuracy during discrete rotation?
- **RQ3:** Do **Audio Source Anchoring** and **Pitch Modulation** introduce additional cognitive cost relative to the no-audio baseline?

4.2 Implementation

The experimental system was implemented in Unity and deployed on a Meta Quest 2 head-mounted display (1832×1920 per eye, 90 Hz refresh rate). The system integrates four key components: (1) an audio beacon positioned at the front of the participant at the start of each trial, (2) a beacon controller that manages audio emission and repositioning based on experimental condition, (3) a snap-turn locomotion interface that triggers beacon repositioning, and (4) real-time pitch modulation synchronized with the rotation transition. The beacon was positioned in front of the participant, selected from four canonical positions (front, back, left, right) evaluated in a pilot study. While personalized head-related transfer functions (p-HRTFs) provide superior spatial localization, they require per-participant calibration that would add experimental overhead and reduce replicability, so we used generic HRTFs to allow direct comparison across participants [Kaul and Rohs 2017]. Front and back produced the most salient directional shifts, but front-back discrimination was unreliable — a known limitation of generic HRTF rendering [Wenzel et al. 1993] — making the frontal position the more perceptually interpretable choice. Participants wore active noise-cancelling headphones to deliver spatial audio and attenuate external sound. Participants were seated in a swivel chair, which allowed physical rotation for the orientation judgment phase.

4.2.1 Virtual Environment. The virtual environment consisted of a room-scale space with a neutral, featureless floor and multiple objects arranged at varying distances and orientations from the participant's starting position (Figure 2). Adapted from Zielasko et al. [Zielasko et al. 2022], floor textures and lighting reflections were removed to prevent visual features from implicitly conveying directional information. Target objects were distributed across 360° around the participant at radii of 3.5, 5.25, and 7 meters, arranged in concentric rings to ensure even distribution across the visual field.

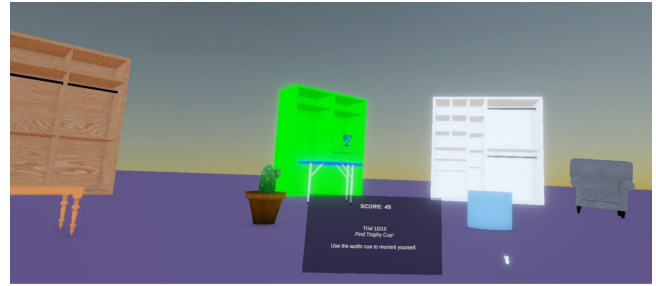


Figure 2: The virtual environment used in the study. Visual features that encode directional information, including floor textures and lighting reflections, were removed to prevent participants from using visual cues for spatial orientation. Target objects were distributed across 360° around the participant at radii of 3.5, 5.25, and 7 meters.

4.2.2 Rotation Method. We implemented joystick-based discrete rotation (snap turning) as the locomotion technique. Participants pushed the thumbstick on the Meta Quest Touch Controller left or right to initiate an immediate 30° discrete rotation in the indicated direction. The 30° increment was chosen to balance rotational granularity and task efficiency, consistent with prior work on discrete rotation increments in VR [Sargunam and Ragan 2018]. To prevent accidental repeated rotations, a cooldown period of 0.4 seconds was enforced between successive rotation commands, maintaining a purely discrete locomotion paradigm. This value was determined through pilot testing: shorter cooldowns allowed unintended multi-step rotations from a single sustained thumbstick push, whereas longer cooldowns made successive turns feel sluggish and disrupted task flow. A 0.4-second interval reliably registered each rotation as a distinct, intentional command while preserving responsive control.

Because the snap turn itself is instantaneous, “rotation transition” refers to the audio playback window triggered by the snap, not the snap motion. In the *Fixed Beacon* conditions, the beacon played a single non-looping pulse triggered on each snap turn (clip duration ≈ 1 s); pitch was held at $1.0\times$ for *Constant* and ramped from $\text{pitchStart} = 1.0$ to $\text{pitchEnd} = 2.5$ over the clip length for *Increasing*, with $t_{01} = (\text{time since clip onset})/(\text{clip length})$. In the *Moving Beacon* conditions, the beacon orbited the participant at $\text{beaconRadius} = 16, \text{units}$ and $\text{snapFollowSpeed} = 25, \text{units/s}$ following a 0.2s post-snap delay; audio looped for the arc duration, with t_{01} normalized over that duration in *Increasing* mode. *Constant* pitch in the moving case used $\text{movingPitch} = 1.0$. The snap turn magnitude was fixed at 30° .

4.3 Participants

We recruited 20 participants (9 male, 11 female; ages 21–34, $M = 25.2$, $SD = 4.4$; 19 right-handed). Participants reported normal or corrected-to-normal hearing and vision with no exclusion criteria based on prior VR experience; VR experience ranged from 0 to over 100 hours ($M = 15$ hrs, $SD = 15$), gaming experience ($M = 80$ hrs, $SD = 26$), and 3D modeling experience ($M = 27$ hrs, $SD = 35$). The study was approved by the Institutional Review Board at the

authors' institution. All participants provided written informed consent prior to participation.

4.4 Task

Each condition consisted of a trial block comprising two phases: a search phase followed by an orientation judgment phase.

Search Phase. At the start of each trial block, the participant was shown a cyan cube marking their starting position and orientation. The cube was then hidden, and a series of 10 target objects were sequentially highlighted in the virtual environment. The participant used 30° snap turns to search for each highlighted target. To select a target, the participant gazed at the object for 3 seconds; a line indicator provided visual feedback of gaze alignment. Upon successful selection, the participant's score was updated. If the participant did not locate the target within 10 seconds, the trial advanced automatically. Translational movement was not allowed — participants could only rotate in place via snap turns. The search phase served as a disorientation task, requiring multiple rotations across varying directions and amplitudes before the orientation judgment.

Orientation Judgment Phase. After completing the search-and-select targets, all target objects and any visual indicators that could disambiguate heading (e.g., landmarks or previously visited target locations) were hidden, leaving a visually uniform environment. The participant was then asked to physically rotate on the swivel chair to face the direction in which they believed their starting position lay, relying solely on internally maintained spatial representations. No feedback on accuracy was provided. The unsigned angular error between the participant's indicated heading and the true direction of the starting position was recorded. This body-based heading judgment follows established spatial updating paradigms in which participants physically orient toward a remembered origin after their viewpoint changed, and has been shown to be a sensitive and valid measure of spatial orientation performance [Lee et al. 2023; Shayman et al. 2024b; Waller and Greenauer 2007; Waller et al. 2004]. Each condition comprised 10 search-and-select targets and 1 orientation judgment.

4.5 Measures

We quantified spatial orientation accuracy using angular error (degrees): the absolute angular deviation between the participant's indicated heading and the true direction of the starting position, computed post-trial. Lower values indicate more accurate spatial updating. Subjective measures included the NASA Task Load Index (TLX), administered after each condition across six subscales using raw scores without pairwise weighting; a condition ranking from most (1) to least preferred (5); a 7-point Likert perception questionnaire covering overall liking, ease of interaction, perceived spatial orientation, perceived disorientation (reverse-coded), spatial awareness confidence, perceived task efficiency, and motion sickness (reverse-coded); open-ended free-response questions on each condition's strengths, weaknesses, and effect on spatial orientation; and the Fast Motion Sickness Scale (FMS) administered before and after each condition on a 0–20 scale.

4.6 Procedure

Each session lasted approximately one hour. Upon arrival, participants provided written informed consent and completed a demographic questionnaire, then donned the Meta Quest 2 headset and noise-cancelling headphones and were seated in a swivel chair. Each condition followed the same sequence: a pre-condition FMS, a brief familiarization phase with the current condition's audio feedback (training data excluded from analysis), the task block, and post-condition FMS and NASA-TLX. After all five conditions, participants completed the condition ranking, perception questionnaire, and free-response questions.

5 Results

5.1 Spatial Orientation Accuracy

Spatial orientation accuracy, measured through angular error Figure 3, varied substantially across conditions, indicating a strong effect of audio design on spatial orientation performance. A one-way ANOVA across all five conditions revealed a significant main effect ($F(4, 76) = 5.975, p < 0.00031, \eta^2 = 0.239$). A 2×2 ANOVA across the four audio conditions confirmed significant main effects of both **Audio Source Anchoring** ($F(1, 19) = 12.836, p < 0.00199, \eta^2 = 0.403$) and **Pitch Modulation** ($F(1, 19) = 5.013, p < 0.0373, \eta^2 = 0.209$). *Fixed + Increasing* achieved the lowest mean angular error ($M = 44.866^\circ$) and *Moving + Constant* the highest ($M = 111.356^\circ$), with *Fixed + Constant* (71.3°), *Moving + Increasing* (90.757°), and *No Audio* (96.8°).

Post hoc pairwise comparisons, reveal that *Fixed + Increasing* produced significantly lower error than three other conditions: *Moving + Constant* ($\Delta = -66.49, p < .001$), *No Audio* ($\Delta = -51.78, p = .009$), and *Moving + Increasing* ($\Delta = -45.89, p = .010$), indicating that *Fixed* audio presentation with increasing pitch is the most effective for heading estimation. The remaining pairwise comparisons did not reach significance (all $p > .200$), suggesting that *Moving + Constant*, *Moving + Increasing*, *Fixed + Constant*, and *No Audio* produce comparable error levels. Notably, anchoring alone (*Fixed + Constant*) did not significantly outperform moving-beacon conditions, implying that the benefit of the *Fixed* design is realized only when combined with increasing pitch.

Critically, *Moving + Constant* performing worse than *No Audio* suggests that a following beacon may not merely fail to support spatial orientation but could introduce a conflicting spatial signal, though this comparison did not reach significance and warrants confirmation. That *Fixed + Increasing* is the only condition to significantly outperform the baseline further suggests that restoring spatial orientation under discrete rotation requires both a stable external auditory reference and a temporal cue to rotational magnitude.

5.2 Subjective Workload (NASA-TLX)

Overall raw TLX did not differ significantly across conditions (Friedman $\chi^2(4) = 3.19, p = .527$; ART ANOVA $F(4, 80) = 1.06, p = .383$). To test the contributions of the two design components, we analyzed each subscale with a 2×2 (Beacon × Pitch) within-subjects ART ANOVA. Only one subscale showed a significant effect (Figure 4).

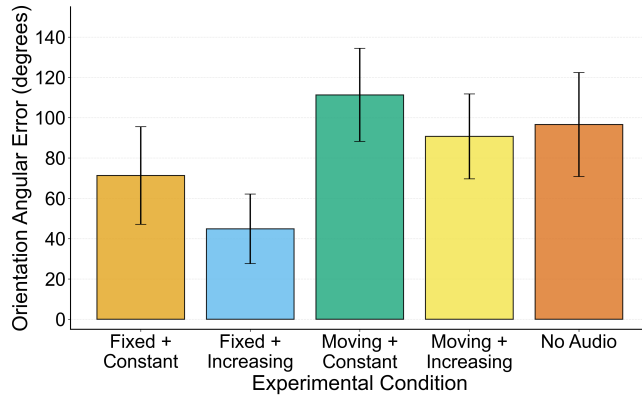


Figure 3: Mean angular error (degrees) across five experimental conditions ($N = 20$). Lower values indicate better spatial orientation accuracy. Error bars indicate 95% confidence intervals.

Temporal Demand. The ART ANOVA revealed a significant main effect of Pitch Modulation, $F(1, 19) = 4.86$, $p = .041$, $\eta_p^2 = .20$, with increasing pitch raising temporal demand from $M = 22.4$ to $M = 29.3$. Neither the Beacon main effect ($F(1, 19) = 0.00$, $p = 1.0$) nor the interaction ($F(1, 19) = 1.17$, $p = .294$) was significant. Descriptively, *Fixed + Increasing* — the condition with the lowest mean angular error — also carried the highest temporal demand ($M = 31.6$, $SE = 6.7$). Post-hoc Wilcoxon tests with Bonferroni correction ($\alpha = .05/6 = .0083$) showed the largest pairwise difference between *Fixed + Constant* and *Fixed + Increasing* ($W = 25.5$, $p = .016$), which did not survive correction.

Effort. Effort was descriptively higher under *Moving* beacons ($M = 35.9$) than *Fixed* beacons ($M = 30.0$), but the ART ANOVA found no significant main effect of Beacon ($F(1, 19) = 3.03$, $p = .099$), Pitch ($F(1, 19) = 0.05$, $p = .823$), or interaction ($F(1, 19) = 0.43$, $p = .521$). The direction is consistent with the interpretation that a spatially unstable auditory source demands additional monitoring effort without yielding useful orientation information, but the effect did not reach significance.

5.3 Motion Sickness (FMS)

FMS scores were near floor across all conditions ($M \leq 0.89$, $Mdn = 0$ on a 0–20 scale), confirming that discrete rotation with spatial audio feedback did not induce meaningful simulator sickness. A Friedman test revealed no significant differences across conditions, $\chi^2(4) = 3.09$, $p = .543$. *Fixed + Constant* Pitch showed the highest mean FMS ($M = 0.89$, $SD = 2.75$), driven primarily by a single outlier (P8, FMS = 12).

5.4 Preference Rankings

Participants ranked the five conditions from most (1) to least (5) preferred after completing all conditions ($N = 20$). *Fixed + Constant* Pitch was most preferred ($M = 2.58$, $Mdn = 2$), followed by *Fixed + Increasing* Pitch ($M = 2.84$, $Mdn = 3$), *Moving + Constant* Pitch ($M = 2.95$, $Mdn = 3$), *Moving + Increasing* Pitch ($M = 3.32$, $Mdn = 3$), and No Audio ($M = 3.32$, $Mdn = 4$). A Friedman test showed no

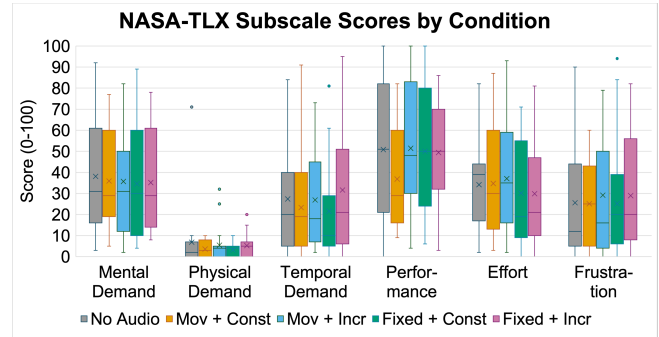


Figure 4: NASA Raw TLX subscale scores ($N = 20$). Pitch and Beacon effects appear on Temporal Demand and Effort; overall TLX is not significant.

significant difference in rankings, $\chi^2(4) = 3.07$, $p = .546$. No Audio was the most polarizing condition (5 first-place and 7 last-place votes), while *Fixed + Increasing* was the most consistently preferred (4 first-place, 1 last-place). Subjective preference favors the comfort of a constant tone, even though the increasing pitch produced the most accurate heading judgments, a dissociation between perceived and actual orientation support that designers must explicitly choose between.

5.5 Perception Ratings

Table 1 summarizes the 7-point Likert perception ratings.

Perceived spatial orientation showed a highly significant effect of condition, $\chi^2(4) = 30.59$, $p < .001$, $W = 0.402$. Post-hoc Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = .05/10 = .005$) revealed that No Audio was rated significantly lower than all four audio conditions: vs. *Fixed + Constant* ($W = 0$, $p_{\text{Bonf}} = .003$), vs. *Moving + Constant* ($W = 0$, $p_{\text{Bonf}} = .009$), vs. *Fixed + Increasing* ($W = 17$, $p_{\text{Bonf}} = .015$), and vs. *Moving + Increasing* ($W = 11.5$, $p_{\text{Bonf}} = .032$). No significant differences were found among the four audio conditions (Friedman 4-condition: $\chi^2(3) = 2.42$, $p = .490$).

Perceived task efficiency showed a marginal effect, $\chi^2(4) = 8.61$, $p = .072$, $W = 0.113$, with *Fixed + Constant* Pitch rated highest ($M = 4.84$) and No Audio lowest ($M = 3.79$). The 2×2 paired t -test showed a marginal Beacon effect, $t(19) = -1.75$, $p = .098$, suggesting that fixed beacon conditions may be perceived as more task-efficient.

Fixed + Constant Pitch was rated highest on overall liking, ease of interaction, spatial orientation, spatial confidence, and task efficiency (Table 1). No Audio was rated lowest on spatial orientation ($M = 2.89$) and spatial confidence ($M = 3.47$). Disorientation ratings did not differ significantly across conditions. Motion sickness ratings were near floor ($M \leq 1.37$), with no meaningful differences.

Notably, *Fixed + Increasing* achieved the best objective orientation accuracy yet ranked second in subjective preference — a dissociation between perceptual effectiveness and user comfort

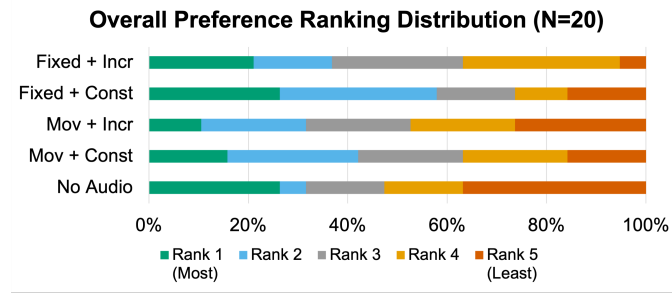


Figure 5: Distribution of preference ranks (1 = most preferred, 5 = least) across the five conditions ($N = 20$). No Audio is bimodal (5 first-place and 7 last-place votes), reflecting an individual-differences split between participants who relied on snap-turn counting and those who attempted audio-based updating. Fixed + Increasing has the most concentrated preference (4 first-place, 1 last-place).

Table 1: Perception questionnaire ratings by condition (7-point Likert scale, Mean $\pm$ SE, $N = 20$). $\downarrow$ indicates reverse-coded items where lower scores are better. Bold indicates the best-performing condition per item.

Item	No Audio	Mov + Const	Mov + Incr	Fixed + Const	Fixed + Incr	Friedman p
Overall Liking	3.95 $\pm$ 0.46	5.05 $\pm$ 0.31	4.37 $\pm$ 0.38	5.26$\pm$0.33	4.58 $\pm$ 0.39	.397
Ease of Interaction	4.11 $\pm$ 0.50	4.68 $\pm$ 0.38	4.79 $\pm$ 0.36	5.00$\pm$0.31	4.84 $\pm$ 0.35	.809
Spatial Orientation	2.89 $\pm$ 0.36	4.84 $\pm$ 0.27	4.79 $\pm$ 0.32	5.32$\pm$0.24	4.84 $\pm$ 0.27	<.001
Disorientation $\downarrow$	4.05 $\pm$ 0.53	3.84$\pm$0.41	3.89 $\pm$ 0.38	4.16 $\pm$ 0.41	4.11 $\pm$ 0.38	.702
Spatial Confidence	3.47 $\pm$ 0.51	3.79 $\pm$ 0.39	3.47 $\pm$ 0.39	4.05$\pm$0.35	3.58 $\pm$ 0.40	.352
Task Efficiency	3.79 $\pm$ 0.46	4.26 $\pm$ 0.35	4.00 $\pm$ 0.40	4.84$\pm$0.30	4.42 $\pm$ 0.40	.072
Motion Sickness $\downarrow$	1.00$\pm$0.00	1.21 $\pm$ 0.16	1.37 $\pm$ 0.22	1.26 $\pm$ 0.17	1.32 $\pm$ 0.22	.166

that the Discussion examines in terms of the attentional cost of temporal auditory encoding.

The critical observation is the participants rated Moving + Constant as similarly orienting to Fixed + Increasing despite a 59° accuracy gap, and rated No Audio as the least orienting condition despite its objective parity with Fixed + Constant. Self-report tracked the *presence* of audio rather than its informational content, indicating that participants cannot reliably introspect on whether an auditory cue is anchored to the world or merely co-moving with them. Subjective measures alone are insufficient for evaluating spatial-audio orientation aids.

5.6 Post-Study Free Responses

Fixed beacon preferred over moving. Ten participants described the fixed beacon as a stable reference, an “audio landmark” (P12: “being in a real room where I know where stuff is”), while eight described moving audio as confusing or indistinguishable from silence (P1: “not useful in orienting myself, so it was more guessing”). Several explicitly noted that the moving beacon offered no external reference, since the sound traveled with them.

Pitch divided. Opinions on pitch split along the accuracy–cost axis. P11 valued the increasing ramp for supplying both “direction... and how much more,” but P4 noted that “the increasing pitch took longer to play than it did to actually snap turn, making it confusing if going too fast,” and P17 reported “focusing more on the sound than where I was.” Constant pitch was broadly described as easier

to process; P7 said it “allowed me to perform tasks faster because I could make an initial movement and quickly react.”

Front-back confusion. Participants reliably distinguished left from right but reported severe front–back confusion (P2: “I can tell whether the audio was from left or right, but not from front or back”), the canonical signature of generic, non-individualized HRTFs, imposing a perceptual ceiling on every audio condition.

Snap-turn counting. Half of participants reported mentally counting snap turns to estimate heading (P19: “okay, I’m now 90° to the right of where I initially was”). Counting persisted even when audio was present, indicating that participants did not promote the auditory cue to a primary heading source. This indicates that participants treated the auditory cues as supplementary rather than primary orientation sources.

Audio–search interference. Six participants reported that attending to audio cues slowed visual search (P9: “I can only focus on one task at a time... I cannot focus on the search task when I am listening to audio”), and six found the tone itself unpleasant.

6 Discussion

6.1 Spatial Stability as a Necessary Condition for Auditory Spatial Updating

The most theoretically significant pattern is not that fixed beacons outperformed moving beacons, but that *Moving + Constant* produced the highest mean angular error of all conditions — descriptively worse than no audio at all, though this difference did not reach statistical significance. This asymmetry suggests that

the benefit of spatial audio in discrete rotation comes not from directional guidance but from providing a stable external reference against which the listener can register self-rotation. A following beacon cancels this shift entirely, conveying no information about the rotation that occurred while introducing an auditory event that may compete with rather than complement the spatial updating process. This interpretation extends Shayman et al.'s [Shayman et al. 2024b] finding that only a fixed beacon improved homing accuracy to discrete rotational updates, where the reference frame must be re-registered after each instantaneous viewpoint change.

6.2 Pitch Modulation as a Temporal Code for Rotational Magnitude

That increasing pitch consistently reduced angular error regardless of beacon type points to a second perceptual mechanism: temporal encoding of rotational magnitude. An upward pitch ramp synchronized with the snap turn may function as an auditory substitute for the missing angular-magnitude signal — its contour providing a graded cue to rotational progression, consistent with sonification principles for monotonic scalar encoding [Dubus and Bresin 2013] and with evidence that temporally continuous sonification during rotation improves spatial accuracy [Ahmetovic et al. 2023]. However, increasing pitch elevated physical and temporal demand, and participants reported temporal misalignment with rapid snap turns. This accuracy-comfort dissociation reflects a fundamental tension in auditory spatial encoding: richer temporal structure carries more rotational information but competes with concurrent visual demands. *Fixed + Constant* minimizes perceptual cost while preserving an allocentric reference, making it the better general-purpose choice; *Fixed + Increasing* is appropriate when orientation accuracy is paramount.

6.3 Perceptual Ceiling and Compensatory Strategies

Even the best condition retained substantial error, pointing to a perceptual ceiling from front-back confusion — the canonical signature of generic non-individualized HRTFs [Wenzel et al. 1993] — which constrained orientation accuracy independently of beacon design. Half of participants adopted snap-turn counting as a compensatory strategy and continued counting even when audio was present, suggesting audio was treated as supplementary rather than primary. Individualized HRTFs, cue reliability feedback, or training may be necessary to shift this balance and allow the perceptual mechanisms identified here to support primary spatial updating.

7 Limitations and Future Work

The findings reported here are specific to 30° snap turn increments and a single virtual environment; generalizability to other increment sizes, continuous rotation, or teleportation-based locomotion remains to be established. Future work could also examine whether the perceptual mechanisms identified here, spatial stability and temporal magnitude encoding, extend to other navigation paradigms, such as multiscale navigation interfaces [Agarwal et al. 2026; Lee et al. 2022; Lee and Stuerzlinger 2025], where users face different demands on maintaining spatial orientation. Each condition comprised a single orientation judgment per participant, which limits

the statistical reliability of the angular error measure; future studies should include multiple trials per condition to increase measurement precision.

Because snap turns are widely adopted in commercial VR as a comfort technique [bcu 2023; Benda et al. 2023; Farmani and Teather 2020], the orientation cost identified here affects a meaningful share of VR users, and likely extends to other interfaces using discrete viewpoint updates, such as stepped rotation in teleportation-based locomotion.

Even in the best-performing condition (*Fixed + Increasing*, $M = 50.9$), errors remain substantial, consistent with angular errors reported in comparable spatial updating paradigms — triangle completion and homing tasks typically yield errors of 30°–60° even under favorable conditions [Shayman et al. 2024b; Waller et al. 2004]. Part of this ceiling reflects generic, non-individualized HRTFs, which caused the front-back confusion participants reported across all conditions independent of beacon design; a sound-localization pre-screening or individualized HRTFs in future work may reveal stronger effects of audio source anchoring than observed here. The virtual environment was also deliberately stripped of directional visual features to isolate audio effects, consistent with prior discrete rotation studies [Zielasko et al. 2022]; future work should test whether these benefits generalize to richer visual environments where audio competes with visual landmarks.

With $N=20$, the within-subjects design and large effect sizes gave sufficient power for the main effects reported, but not for smaller anchoring $\times$ pitch interaction effects; future work with larger samples should test these interactions across task conditions and snap turn increments.

8 Conclusion

This work investigated whether and how spatial audio feedback can restore the perceptual signals disrupted by discrete rotation in VR — a question prior work had not directly addressed. Our findings reveal two distinct perceptual mechanisms through which spatial audio can support spatial orientation under snap turns, and one fundamental condition that determines whether it helps or harms.

First, spatial stability is a necessary condition for auditory spatial updating: a world-fixed beacon supplies an allocentric reference that partially substitutes for the optic flow eliminated by discrete rotation, whereas a following beacon cancels this shift and may interfere with rather than support orientation.

Second, pitch modulation functions as a temporal code for rotational magnitude, encoding angular displacement as a gradually changing pitch and enabling more accurate heading reconstruction. This benefit comes at a measurable attentional cost, elevating temporal and physical demand, so *Fixed + Constant* is the better general-purpose design while *Fixed + Increasing* suits applications where orientation accuracy is paramount.

Together, these findings establish that the perceptual utility of spatial audio under discrete rotation depends on spatial stability, and that temporal pitch can serve as an auditory proxy for rotational cues, with implications for any VR interface that employs discontinuous viewpoint transformations.

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